

Jens Robben

Roetersstraat 11, 1018 WB Amsterdam, the Netherlands

✉ j.robben@uva.nl | 🌐 [Personal website](#) | 🔗 [LinkedIn](#)

RESEARCH INTERESTS

In general: Actuarial Science, Data Science, Machine Learning, Applied Statistics

In particular: Life Insurance, Mortality Modelling, Climate Risk, Non-Life Insurance, Reserving

ACADEMIC POSITIONS

University of Amsterdam 2024 - Present

Postdoctoral researcher in Actuarial Science

KU Leuven 2025 - Present

Part-time lecturer in Actuarial Science

EDUCATION

KU Leuven 2020 - 2024

PhD candidate in Insurance

Supervisor: Katrien Antonio

University of Amsterdam September 2023

Visiting scholar

Host: Torsten Kleinow

KU Leuven 2019 - 2020

MSc in Actuarial and Financial Engineering - *Summa cum laude*

KU Leuven 2017 - 2019

MSc in Mathematics - *Summa cum laude*

KU Leuven 2014 - 2017

BSc in Mathematics - *Magna cum laude*

PUBLICATIONS

J. Robben, K. Barigou, & T. Kleinow (2026). [Granular mortality modelling with temperature and epidemic shocks: a three-state regime-switching approach](#). *Insurance: Mathematics and Economics*, 103250.

J. Robben, K. Antonio, & T. Kleinow (2026). [The short-term association between environmental variables and mortality: evidence from Europe](#). *Journal of the Royal Statistical Society Series A: Statistics in Society*, 189(2), 1131-1153.

J. Robben & K. Antonio (2024). [Catastrophe risk in a stochastic multi-population mortality model](#). *Journal of Risk and Insurance*, 91, 599–651.

J. Crevecoeur, J. Robben, & K. Antonio. (2022). [A hierarchical reserving model for reported non-life insurance claims](#). *Insurance: Mathematics and Economics*, 104, 158-184.

J. Robben, K. Antonio, & S. Devriendt. (2022). [Assessing the impact of the COVID-19 shock on a stochastic multi-population mortality model](#). *Risks*, 10(2), 26, 1-33.

WORKING PAPERS

J. Robben & K. Barigou (2025). [A penalized distributed lag non-linear Lee-Carter framework for regional weekly mortality forecasting](#). arXiv preprint arXiv:2509.24087.

→ Under revision at the *International Journal of Forecasting*.

J. Robben & T. Kleinow (2026). [Socio-demographic inequalities in the maximum human lifespan](#). arXiv preprint arXiv:2604.07974.

→ Under revision at *PLOS One*.

K. So, K. Barigou, & J. Robben. (2026). [Climate-driven mortality forecasting using deep learning](#). arXiv preprint arXiv:2606.26980.

→ Submitted.

WORK IN PROGRESS

A Dutch administrative index of socioeconomic heat inequality

with Andrew Cairns (Heriot-Watt University) and Torsten Kleinow (University of Amsterdam).

Environmental determinants of cause-specific mortality across socio-economic groups in Australia

with Han Li (University of Melbourne), Séverine Arnold (University of Lausanne), and Tim Adair (University of Melbourne).

Associating socio-economic, demographic, and environmental factors with mortality improvements: a cause-of-death study in Europe

with Séverine Arnold (University of Lausanne) and Eman ElMeaddawy (University of Lausanne).

A multi-peril framework for motor insurance claims: unraveling risk drivers of electrified vehicles

with Katrien Antonio (KU Leuven) and Dries Van Ceulebroeck (KU Leuven).

CONFERENCES & SEMINARS

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| 2026 | 33rd International Congress of Actuaries (Tokyo)*, 29th International Congress on Insurance: Mathematics and Economics (Seoul), Scandinavian Actuarial Conference (Stockholm), 1st ASTIN Bulletin Conference (Zurich) |
| 2025 | MLISTRAL 2 (Marseille), Working Group on Risk - CREAM (Paris), Climate Change and Insurance (Edinburgh), 28th International Congress on Insurance: Mathematics and Economics (Tartu), Actuarial, Finance, Risk and Insurance Congress (Kenya) [†] , Insurance Data Science Conference (London), Webinar for the International Actuarial Association - Life Section (online) [†] , Gompertz 200 (Amsterdam), Doctoral school seminar ULB (Brussels) [†] |
| 2024 | Longevity 19 (Amsterdam), Climate Change and Insurance (Vienna), Webinar for the International Actuarial Association - Life Section (online), 27th International Congress on Insurance: Mathematics and Economics (Chicago), Insurance Data Science Conference (Stockholm) [†] , Climate change seminar - Actuarial society of South Africa (online) [†] , KU Leuven, Bayes, and UvA PhD Workshop (Leuven) |
| 2023 | Longevity 18 (London), Modelling and Societal Impact of Longevity and Ageing Conference (Amsterdam), 26th International Congress on Insurance: Mathematics and Economics (Edinburgh), Perspectives on actuarial risks in talks of young researchers (Valencia) [†] , RCLR seminar (Amsterdam), Doctoral Seminar (Leuven) |
| 2022 | Insurance Data Science Conference (Milan), European Actuarial Journal Conference (Tartu), Doctoral Seminar (Leuven), Presentation at AG Insurance Campus (Brussels) [†] |
| 2021 | 24th International Congress on Insurance: Mathematics and Economics (online) |

(* indicates scheduled, [†] indicates by coauthors)

GRANTS & AWARDS

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| 2026 | Nominee for actuarial talent of the year (Dutch Actuarial Association) |
| 2025 | IAALS Research Grant with Andrew Cairns and Torsten Kleinow (PI, €18,000)
"A Machine Learning Approach to Constructing a Socio-Economic Heat Vulnerability Index Using Individual-Level Mortality Data" |
| 2025 | Winner of the Netspar PhD Thesis Award (€3,000) |
| 2020 | AFI Master Thesis Award (KU Leuven, €500) |

TEACHING EXPERIENCE

2025-	Loss Models (MSc, KU Leuven)	Lecturer
2025-	Advanced Life Insurance Mathematics (MSc, KU Leuven)	Lecturer
2024-	Life Insurance Mathematics (BSc, UvA)	TA for Katrien Antonio
2024-	Introduction Econometrics & Actuarial Science (BSc, UvA)	TA for Frank van Berkum
2024-	Introduction Data Science (BSc, UvA)	TA for Daniël Linders
2024-	Master's Theses in Actuarial Science (MSc, UvA)	Daily supervisor
2024-	Bachelor's Theses in Actuarial Science (BSc, UvA)	Daily supervisor
2020-2024	Loss Models (MSc, KU Leuven)	TA for Katrien Antonio
2020-2024	Master's Theses in Actuarial and Financial Engineering (MSc, KU Leuven)	Daily supervisor
2024	Modelling and Quantifying Mortality and Longevity Risk (35th International Summer School of the Swiss Association of Actuaries)	Lecturer

PROFESSIONAL SERVICE

Referee	Insurance: Mathematics and Economics Astin Bulletin Scandinavian Actuarial Journal Annals of Actuarial Science European Actuarial Journal Scientific Reports Journal of Population Research
External reviewer	Discovery Grants programme, Natural Sciences and Engineering Research Council of Canada (NSERC)
Summer School	Teaching team member, 35th International Summer School on Modelling and quantifying mortality and longevity risk, Swiss Association of Actuaries, University of Lausanne (2024)
Fellow/Member	Research Centre for Longevity Risk, University of Amsterdam (2024-) NETSPAR: Network for Studies on Pensions, Aging and Retirement (2025-) Leuven Research Centre for Insurance and Financial Risk Analysis (2020-2024)

POLICY REPORTS

K. Antonio, S. Devriendt, J. Robben, & D. Sznajder. (2020). Assessing the impact of COVID-19 on the IA|BE 2020 mortality projections: a scenario analysis. *Published by the Institute of Actuaries in Belgium.*

K. Antonio, S. Devriendt, & J. Robben. (2020). [The IA|BE 2020 mortality projection model for the Belgian population](#). *Published by the Institute of Actuaries in Belgium.*

BIBLIOMETRIC INDICATORS

Google Scholar (August 9, 2026)

	All	Since 2021
Citations	121	120
h-index	6	6
i10-index	5	5

SKILLS

Software: R, L^AT_EX, Python

Language: Dutch (native), English (fluent), French (basic)

BIO

Birth: September 7, 1996

Citizenship: Belgian

Hobbies: Running, Hiking